Experiment No. 09

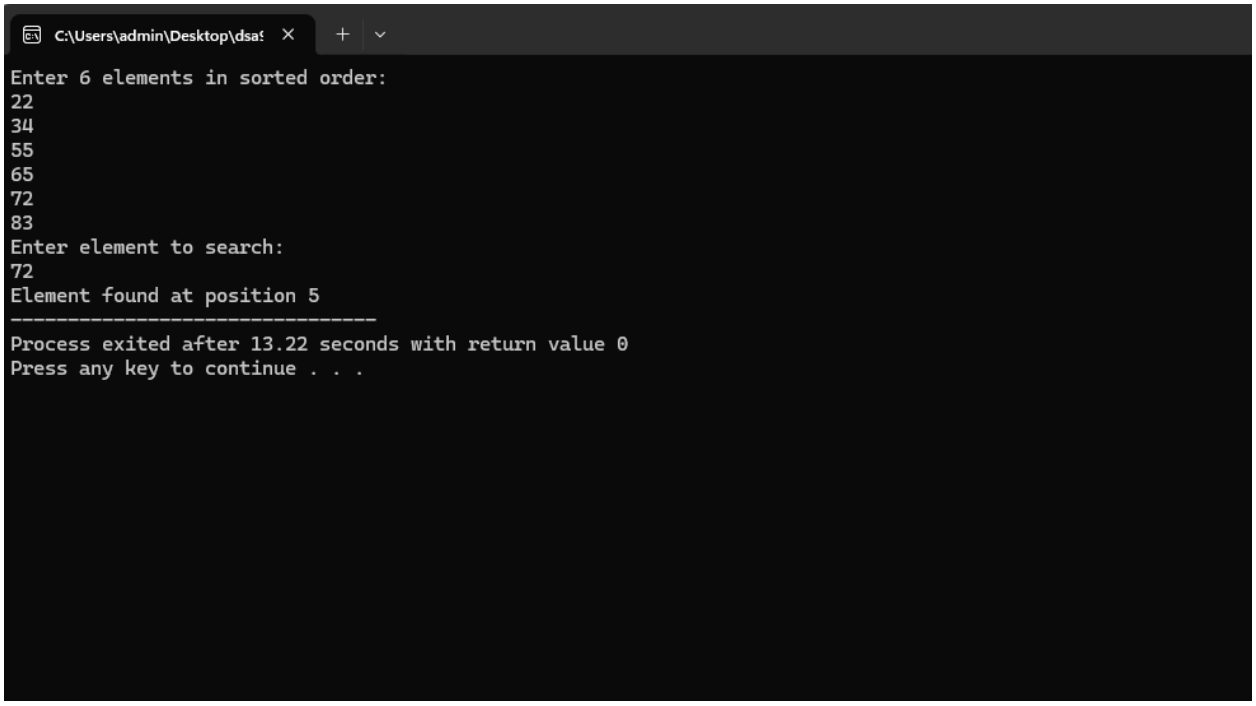
```c
#include<stdio.h>
int main()
{
    int a[6],i,key,mid;
    int low=0,high=5;
printf("Enter 6 elements in sorted order:\n");
 for(i=0;i<6;i++)
 {
    scanf("%d",&a[i]);
 }
 printf("Enter element to search:\n");
 scanf("%d",&key);
 while(low<=high)
 {
    mid=(low+high)/2;

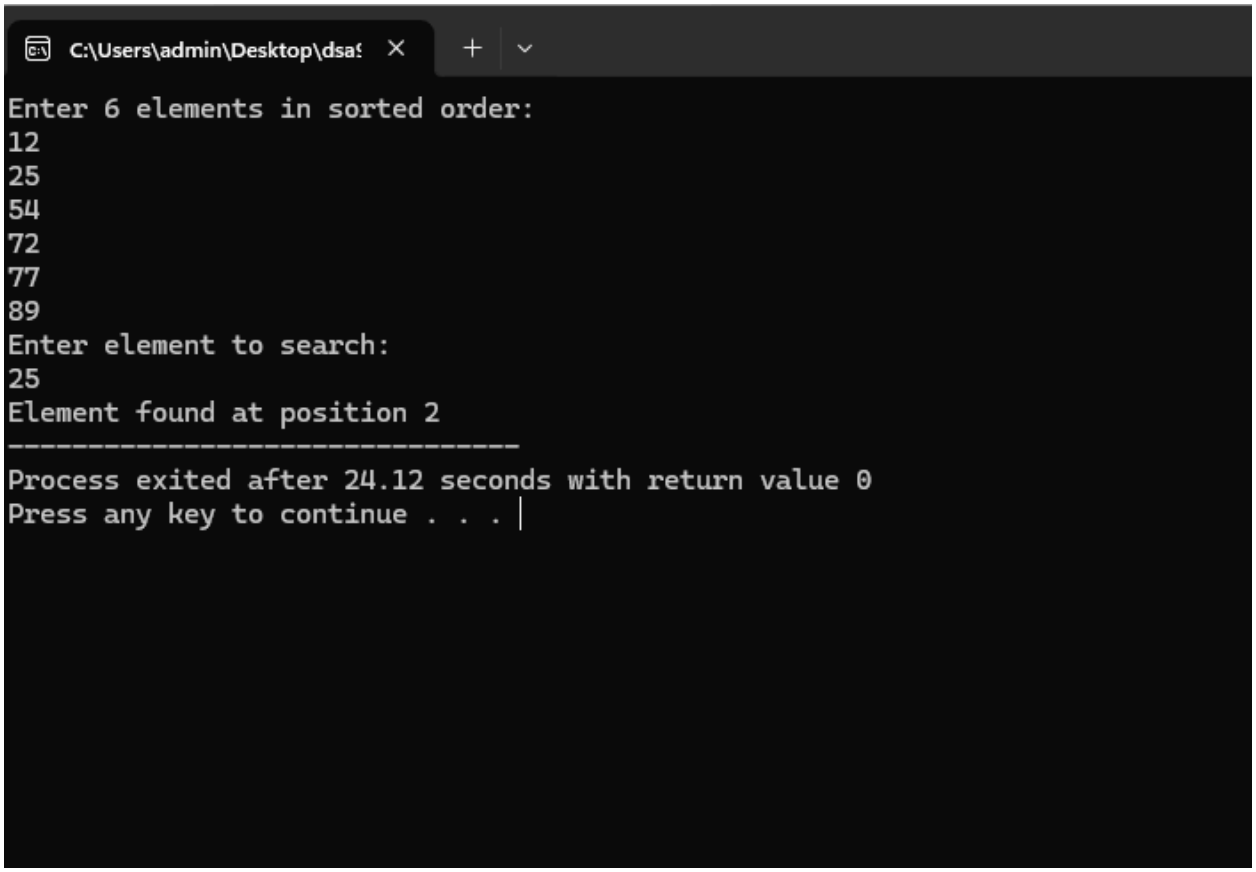
    if(a[mid]==key)
    {
        printf("Element found at position %d",mid+1);
        return 0;
     }
     else if(a[mid]<key)
     {
        low=mid+1;
     }
     else
     {
        high=mid-1;
     }
}
```

```
while(low<=high)
{
    mid=(low+high)/2;

    if(a[mid]==key)
    {
        printf("Element found at position %d",mid+1);
        return 0;
    }
    else if(a[mid]<key)
    {
        low=mid+1;
    }
    else
    {
        high=mid-1;
    }
}
printf("Element not found!!");
return 0;
}
```

OUTPUT:




C:\Users\admin\Desktop\dsa
Enter 6 elements in sorted order:
12
25
54
72
77
89
Enter element to search:
25
Element found at position 2

Process exited after 24.12 seconds with return value 0
Press any key to continue . . .